=== CONFIGURACION GNB (1780311715) ===

```yaml
# gNB configuration for O-RAN E2E V2X scenario
# Based on srsRAN_Project gNB configuration

# Core network configuration
amf_config:
  amf_addr: 10.53.1.2
  amf_port: 38412
  # PLMN ID for the network
  plmn_id:
    mcc: 315
    mnc: 010
  # Tracking Area Code
  tac: 7
  # Slice support list for V2X services
  tai_slice_support_list:
    - sst: 2 # Standardized SST for V2X services (3GPP TS 23.501, Table 5.15.2.2-1)
      sd: 0x000001 # Example SD, can be adjusted
  # gNB ID (20 bits)
  gnb_id: 0x18000
  # gNB name
  gnb_name: srsgnb-v2x

# gNB CU-CP configuration
cu_cp:
  # IP address for N2 interface (gNB to AMF)
  bind_addr: 10.53.1.3
  # Port for N2 interface
  bind_port: 38412

# gNB DU configuration
du:
  # DU ID
  du_id: 0x10000
  # F1-C interface configuration
  f1_c:
    # IP address for F1-C interface
    bind_addr: 10.53.1.3
    # Port for F1-C interface
    bind_port: 8000

# Radio Frequency (RF) configuration
rf:
  # Device type (ZMQ for virtual radio)
  device_name: zmq
  # Device arguments for ZMQ
  device_args: "tx_port=tcp://*:2000,rx_port=tcp://0.0.0.0:2001,id=gnb,base_srate=23.04e6"
  # Transmit gain
  tx_gain: 70
  # Receive gain
  rx_gain: 30
  # Base sample rate
  srate: 23.04e6
  # Number of antennas
  nof_antennas: 1

# NR (New Radio) configuration
nr:
  # NR band
  bands: [3]
  # Channel bandwidth in MHz
  channel_bandwidth_MHz: 20
  # Downlink NR ARFCN
  dl_nr_arfcn: 368500
  # SSB NR ARFCN
```

```yaml
  ssb_nr_arfcn: 367930
  # Number of physical resource blocks
  nof_prb: 106
  # Maximum number of physical resource blocks
  max_nof_prb: 106
  # Cell ID
  cell_id: 1
  # Physical Cell ID
  phy_cell_id: 1
  # Subcarrier spacing (kHz)
  ssb_subcarrier_spacing: 30
  # SSB periodicity (ms)
  ssb_periodicity: 20
  # TDD configuration
  tdd_config:
    # Duplex mode
    duplex_mode: tdd
    # Slot configuration (DL/UL ratio)
    slot_config:
      - { dl_symbols: 8, ul_symbols: 2, guard_symbols: 4 } # Example: 8 DL, 2 UL, 4 Guard
    # Frame structure
    frame_structure:
      # Number of slots per frame
      nof_slots_per_frame: 10
      # Number of slots per subframe
      nof_slots_per_subframe: 1
      # Number of symbols per slot
      nof_symbols_per_slot: 14

# Logging configuration
log:
  # Log level for all modules
  all_level: info
  # Log level for PHY library
  phy_lib_level: none
  # Hex dump limit
  all_hex_limit: 32
  # Log filename
  filename: /tmp/gnb.log
  # Max log file size (-1 for unlimited)
  file_max_size: -1

# PCAP configuration
pcap:
  # Enable PCAP capture (none, mac, mac_nr, nas)
  enable: none
  # MAC filename
  mac_filename: /tmp/gnb_mac.pcap
  # MAC NR filename
  mac_nr_filename: /tmp/gnb_mac_nr.pcap
  # NAS filename
  nas_filename: /tmp/gnb_nas.pcap
```

=== CONFIGURACION UE (1780311715) ===

```
# UE configuration for O-RAN E2E V2X scenario
# Based on srsRAN_Project UE configuration

[rf]
# Device type (ZMQ for virtual radio)
device_name = zmq
# Device arguments for ZMQ
device_args = tx_port=tcp://localhost:2001,rx_port=tcp://*:2000,id=ue,base_srate=23.04e6
# Transmit gain
tx_gain = 20
# Receive gain
rx_gain = 10
# Base sample rate
srate = 23.04e6
# Number of antennas
nof_antennas = 1

[rat.nr]
# NR band
bands = 3
# Number of carriers
nof_carriers = 1
# Number of physical resource blocks
nof_prb = 106
# Maximum number of physical resource blocks
max_nof_prb = 106
# Downlink NR ARFCN
dl_nr_arfcn = 368500
# SSB NR ARFCN
ssb_nr_arfcn = 367930

[usim]
# USIM mode (soft for software-based)
mode = soft
# Authentication algorithm
algo = milenage
# Operator Key (OPC)
opc  = 63BFA50EE6523365FF14C1F45F88737D
# Subscriber Key (K)
k    = 00112233445566778899aabbccddeeff
# International Mobile Subscriber Identity (IMSI)
imsi = 315010000000001
# International Mobile Equipment Identity (IMEI) - example
imei = 353490069873319

[nas]
# Access Point Name (APN)
apn = srsapn
# APN protocol
apn_protocol = ipv4

[rrc]
# RRC release version
release = 15
# UE category
ue_category = 4

[gw]
# Gateway configuration (no ip_addr as per strict rule)

[pcap]
# Enable PCAP capture (none, mac, mac_nr, nas)
enable = none
# MAC filename
```

```
mac_filename = /tmp/ue_mac.pcap
# MAC NR filename
mac_nr_filename = /tmp/ue_mac_nr.pcap
# NAS filename
nas_filename = /tmp/ue_nas.pcap

[log]
# Log level for all modules
all_level = info
# Log level for PHY library
phy_lib_level = none
# Hex dump limit
all_hex_limit = 32
# Log filename
filename = /tmp/ue.log
# Max log file size (-1 for unlimited)
file_max_size = -1
```

=== DOCKER COMPOSE (1780311715) ===

```
# Copyright 2020-2022 Open Networking Foundation
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.

version: '3.8'
services:
  5gc:
    build:
      context: .
      dockerfile: Dockerfile-5gc
    container_name: open5gs-5gc
    privileged: true
    command: 5gc -c open5gs-5gc.yml
    volumes:
      - ./open5gs-5gc.yml:/open5gs/install/etc/open5gs/open5gs-5gc.yml
    networks:
      ran:
        ipv4_address: 10.53.1.2
    ports:
      - "3000:3000" # WebUI
      - "38412:38412/sctp" # N2 AMF
      - "2123:2123/udp" # PFCP UPF
      - "80:80" # NRF
      - "9090:9090" # Prometheus
    cap_add:
      - NET_ADMIN
      - SYS_NICE
    healthcheck:
      test: ["CMD", "pgrep", "open5gs-amfd"]
      interval: 10s
      timeout: 5s
      retries: 5
  gnb:
    build:
      context: .
      dockerfile: Dockerfile-gnb
    container_name: srsran-gnb
    privileged: true
    command: srsgnb -c gnb.yaml
    volumes:
      - ./gnb.yaml:/srsran/install/etc/srsran/gnb.yaml
    networks:
      ran:
        ipv4_address: 10.53.1.3
    cap_add:
      - NET_ADMIN
      - SYS_NICE
    healthcheck:
      test: ["CMD", "pgrep", "srsgnb"]
      interval: 10s
      timeout: 5s
      retries: 5

networks:
```

```
ran:
  driver: bridge
  ipam:
    config:
      - subnet: 10.53.1.0/24
```